Introduction to Mathematics for Deep Learning: Final Solutions

1. **Mix of Questions** *(40 points, 10 points each)*

   1) If a convex function f is μ-strongly convex, show that
   $$\nabla^2 f(x) \succeq \mu I.$$

   2) Show that the following two forms of Nesterov AGM are equivalent.
   (Method 1):
   $$\begin{aligned} x_{k+1} =& y_k - \frac{1}{L}\nabla f(y_k) \\ y_{k+1} =& x_{k+1} + \frac{k-1}{k+2}(x_{k+1} - x_k). \end{aligned}$$
   where $x_0 = y_0$.
   (Method 2):
   $$\begin{aligned} x_{k+1} =& y_k - \frac{1}{L}\nabla f(y_k) \\ z_{k+1} =& z_k - \frac{k+1}{2L}\nabla f(y_k) \\ y_{k+1} =& \left(1 - \frac{2}{k+2}\right) x_{k+1} + \frac{2}{k+2} z_{k+1} \end{aligned}$$
   where $x_0 = y_0 = z_0$. All you need to show is that x_k values coincide in both methods.

   3) For μ-strongly convex function f, show that
   $$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta) f(y) - \frac{\mu}{2}\theta(1-\theta)\|x-y\|^2$$
   for all x, y.

   4) Consider the problem $\min_{x \in \mathbb{R}^d} f(x)$ where $f(x)$ is μ-strongly convex and L-smooth. The method
   $$\begin{aligned} x_{k+1} =& y_k - \frac{1}{L}\nabla f(y_k) \\ y_{k+1} =& x_{k+1} + \frac{\sqrt{\nu}-1}{\sqrt{\nu}+1}(x_{k+1} - x_k) \end{aligned}$$
   where $k = 0, 1, \ldots$, $x_0 = y_0$, and $\nu = L/\mu$ is called the strongly convex accelerated gradient method. Suppose we HAVE the following sequence of V_k is decaying
   $$V_k = \left(1 + \frac{1}{\sqrt{\nu}-1}\right)^k (f(x_k) - f(x^\star) + \frac{\mu}{2}\|z_k - x^\star\|^2)$$
   where $z_k = (1+\sqrt{\nu})y_k - \sqrt{\nu}x_k$ (NOTE: you can use the above fact without proof). Show that
   $$f(x_k) - f(x^\star) \leq \frac{\mu + L}{2}\|x_0 - x^\star\|^2 e^{-k/\sqrt{\nu}}.$$

Solution:

1) By the definition of strong convexity, f is μ-strongly convex if for all $x, y \in \mathbb{R}^n$,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|y - x\|^2.$$

To derive the result for the Hessian, consider the second-order Taylor expansion of f at x:

$$f(y) = f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2}(y - x)^T \nabla^2 f(z)(y - x).$$

where z lies in between x and y.
From the definition of μ-strong convexity, substituting the Taylor expansion gives:

$$\frac{1}{2}(y - x)^T \nabla^2 f(z)(y - x) \geq \frac{\mu}{2}\|y - x\|^2.$$

Rearranging, we get:

$$(y - x)^T \nabla^2 f(z)(y - x) \geq \mu\|y - x\|^2, \quad \forall y, x \in \mathbb{R}^n.$$

This inequality implies that the Hessian $\nabla^2 f(z)$ is positive semidefinite and satisfies:

$$\nabla^2 f(x) \succeq \mu I.$$

Thus, the result is proven.

2) (Induction)
Suppose that the sequences x_k and y_k from both methods match up to iteration k.
From Method 2, we have

$$z_{k+1} = z_k + \frac{k+1}{2}(x_{k+1} - y_k).$$

From Method 2, we have

$$\begin{aligned}
y_{k+1} &= \left(1 - \frac{2}{k+2}\right) x_{k+1} + \frac{2}{k+2} z_{k+1} \\
&= \left(1 - \frac{2}{k+2}\right) x_{k+1} + \frac{2}{k+2}\left(z_k + \frac{k+1}{2}(x_{k+1} - y_k)\right) \\
&= x_{k+1} + \frac{k-1}{k+2} x_{k+1} + \frac{2}{k+2} z_k - \frac{k+1}{k+2} y_k \\
&= x_{k+1} + \frac{k-1}{k+2}(x_{k+1} - x_k) + \frac{k+1}{k+2}\left(\frac{k-1}{k+1} x_k + \frac{2}{k+1} z_k - y_k\right) \\
&= x_{k+1} + \frac{k-1}{k+2}(x_{k+1} - x_k).
\end{aligned}$$

where the last inequality is due to the definition of y_k in Method 2. This coincide with the formula in Method 1.

3) Let $z = \theta x + (1-\theta)y$. Then,

$$\begin{aligned} f(x) \geq & f(z) + \langle \nabla f(z), x - z \rangle + \frac{\mu}{2}\|x - z\|^2 \\ = & f(z) + (1-\theta)\langle \nabla f(z), x - y \rangle + \frac{\mu}{2}(1-\theta)^2\|x - y\|^2 \end{aligned}$$

Similarly,

$$\begin{aligned} f(y) \geq & f(z) + \langle \nabla f(z), y - z \rangle + \frac{\mu}{2}\|y - z\|^2 \\ = & f(z) - \theta\langle \nabla f(z), x - y \rangle + \frac{\mu}{2}\theta^2\|x - y\|^2 \end{aligned}$$

Then,

$$\begin{aligned} \theta f(x) + (1-\theta) f(y) \geq & f(z) + \frac{\mu}{2}\theta(1-\theta)\|x - y\|^2 \\ = & f(\theta x + (1-\theta)y) + \frac{\mu}{2}\theta(1-\theta)\|x - y\|^2 \end{aligned}$$

which concludes the proof.

4) We have

$$\begin{aligned} V_k \leq V_0 = & f(x_0) - f(x^\star) + \frac{\mu}{2}\|z_0 - x^\star\|^2 \\ = & f(x_0) - f(x^\star) + \frac{\mu}{2}\|x_0 - x^\star\|^2 \\ = & \frac{L}{2}\|x_0 - x^\star\|^2 + \frac{\mu}{2}\|x_0 - x^\star\|^2 \\ = & \frac{\mu + L}{2}\|x_0 - x^\star\|^2. \end{aligned}$$

On the other hand,

$$\begin{aligned} V_k &= \left(\frac{\sqrt{\nu}}{\sqrt{\nu} - 1}\right)^k (f(x_k) - f(x^\star) + \frac{\mu}{2}\|z_k - x^\star\|^2) \\ &\geq \left(\frac{\sqrt{\nu}}{\sqrt{\nu} - 1}\right)^k (f(x_k) - f(x^\star)) \end{aligned}$$

Thus,

$$\begin{aligned} f(x_k) - f(x^\star) \leq & \left(1 - \frac{1}{\sqrt{\nu}}\right)^k \frac{\mu + L}{2}\|x_0 - x^\star\|^2 \\ \leq & e^{-k/\sqrt{\nu}} \frac{\mu + L}{2}\|x_0 - x^\star\|^2 \end{aligned}$$

where the last inequality is from $1 - x \leq e^{-x}$.

2. **Lyapunov** *(30 points)*
For a L-smooth convex function f and $0 < \theta < 1$, consider the gradient descent problem with slightly changed parametrization.

$$x_{k+1} = x_k - \frac{2\theta}{L}\nabla f(x_k).$$

(a) Using the co-coercivity

$$\langle \nabla f(y) - \nabla f(x), y - x\rangle \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2,$$

show that

$$\|x_{k+1} - x_k\|^2 \leq \|x_k - x_{k-1}\|^2.$$

(10 points)

(b) Let

$$V_k = k\frac{1-\theta}{\theta}\|x_k - x_{k-1}\|^2 + \|x_k - x^\star\|^2$$

where $x^\star$ is the global minimum. Show that $V_{k+1} \leq V_k$. *(10 points)*

(c) Show that

$$\|x_{k+1} - x_k\|^2 \leq \frac{\theta}{1-\theta}\frac{1}{k+1}\|x_0 - x^\star\|^2$$

(10 points)

Solution:

(a) We have

$$\begin{aligned}
\|x_{k+1} - x_k\|^2 =& \|x_k - x_{k-1} - \frac{2\theta}{L}\nabla(f(x_k) - \nabla f(x_{k-1}))\|^2 \\
=& \|x_k - x_{k-1}\|^2 \\
& - \frac{4\theta}{L}\left(\langle \nabla f(x_k) - \nabla f(x_{k-1}), x_k - x_{k-1}\rangle - \frac{\theta}{L}\|\nabla f(x_k) - \nabla f(x_{k-1})\|^2\right) \\
\leq& \|x_k - x_{k-1}\|^2 \\
& - \frac{4\theta}{L}\left(\langle \nabla f(x_k) - \nabla f(x_{k-1}), x_k - x_{k-1}\rangle - \frac{1}{L}\|\nabla f(x_k) - \nabla f(x_{k-1})\|^2\right) \\
\leq& \|x_k - x_{k-1}\|^2.
\end{aligned}$$

(b) We have

$$V_{k+1} - V_k = (k+1)\frac{1-\theta}{\theta}\|x_{k+1} - x_k\|^2 + \|x_{k+1} - x^\star\|^2$$

$$
\begin{aligned}
&- k\frac{1-\theta}{\theta}\|x_k - x_{k-1}\|^2 - \|x_k - x^\star\|^2 \\
\leq &\frac{1-\theta}{\theta}\|x_{k+1} - x_k\|^2 + \|x_{k+1} - x^\star\|^2 - \|x_k - x^\star\|^2 \\
= &\frac{1-\theta}{\theta}\frac{4\theta^2}{L^2}\|\nabla f(x_k)\|^2 + \|x_k - x^\star - \frac{2\theta}{L}\nabla f(x_k)\|^2 - \|x_k - x^\star\|^2 \\
= &\frac{1-\theta}{\theta}\frac{4\theta^2}{L^2}\|\nabla f(x_k)\|^2 + \frac{4\theta^2}{L^2}\|\nabla f(x_k)\|^2 - \frac{4\theta}{L}\langle x_k - x^\star, \nabla f(x_k)\rangle \\
= &\frac{4\theta}{L}\left(\frac{1}{L}\|\nabla f(x_k)\|^2 - \langle x_k - x^\star, \nabla f(x_k)\rangle\right) \\
= &\frac{4\theta}{L}\left(\frac{1}{L}\|\nabla f(x_k) - \nabla f(x^\star)\|^2 - \langle x_k - x^\star, \nabla f(x_k) - \nabla f(x^\star)\rangle\right) \\
\leq &0.
\end{aligned}
$$

where the last inequality is co-coercivity again with $\nabla f(x^\star) = 0$.

(c) Again,

$$
\begin{aligned}
(k+1)\frac{1-\theta}{\theta}\|x_{k+1} - x_k\|^2 \leq &V_{k+1} \\
\leq &V_0 \\
\leq &\|x_0 - x^\star\|^2.
\end{aligned}
$$

Thus,

$$
\|x_{k+1} - x_k\|^2 \leq \frac{\theta}{1-\theta}\frac{1}{k+1}\|x_0 - x^\star\|^2
$$

3. **Operators** *(30 points, 10 points each)*
For $0 < L < \infty$, define the class of L-Lipschitz operators as

$$
\mathcal{L}_L = \{\mathbb{A} : \mathbb{R}^n \to \mathbb{R}^n \mid \|\mathbb{A}x - \mathbb{A}y\|^2 \leq L^2\|x - y\|^2, \text{ for all } x, y\}.
$$

For $0 < \beta < \infty$, define the class of β-cocoercive operators as

$$
\mathcal{C}_\beta = \{\mathbb{A} : \mathbb{R}^n \to \mathbb{R}^n \mid \langle \mathbb{A}x - \mathbb{A}y, x - y\rangle \geq \beta\|\mathbb{A}x - \mathbb{A}y\|^2, \text{ for all } x, y\}.
$$

For $0 < \theta < 1$, define the classs of θ-averaged operators as

$$
\mathcal{N}_\theta = \{(1-\theta)\mathbb{I} + \theta\mathbb{A} \mid \mathbb{A} \in \mathcal{L}_1\}.
$$

(a) For $0 < R < 1$, show that

$$
\mathcal{L}_R \subseteq \mathcal{N}_{\frac{1+R}{2}} \subseteq \mathcal{L}_1.
$$

(b) Show that $\mathbb{A} \in \mathcal{N}_\theta$ if and only if $\mathbb{I} - \mathbb{A} \in \mathcal{C}_{1/(2\theta)}$.

(c) Show that the following four statements are equivalent

i. $\mathbb{A} \in \mathcal{N}_{1/2}$
ii. $\mathbb{A} \in \mathcal{C}_1$
iii. $\mathbb{I} - \mathbb{A} \in \mathcal{N}_{1/2}$
iv. $2\mathbb{A} - \mathbb{I} \in \mathcal{L}_1$

Solution:

(a) Let $\mathbb{A} \in \mathcal{L}_R$, then

$$\mathbb{A} = \left(1 - \frac{1+R}{2}\right)\mathbb{I} + \frac{1+R}{2}\left(\frac{2}{1+R}\mathbb{A} - \frac{1-R}{1+R}\mathbb{I}\right).$$

It is enough to show that

$$\mathbb{B} = \frac{2}{1+R}\mathbb{A} + \frac{1-R}{1+R}\mathbb{I} \in \mathcal{L}_1.$$

We have

$$\begin{aligned}
\|\mathbb{B}x - \mathbb{B}y\| =& \left\| \frac{2}{1+R}\mathbb{A}x - \frac{1-R}{1+R}x - \frac{2}{1+R}\mathbb{A}y + \frac{1-R}{1+R}y \right\| \\
\leq& \left\| \frac{2}{1+R}\mathbb{A}x - \frac{2}{1+R}\mathbb{A}y \right\| + \left\| -\frac{1-R}{1+R}x + \frac{1-R}{1+R}y \right\| \\
\leq& \frac{2}{1+R}R\|x-y\| + \frac{1-R}{1+R}\|x-y\| \\
\leq& \|x-y\|
\end{aligned}$$

which implies $\mathcal{L}_R \subseteq \mathcal{N}_{\frac{1+R}{2}}$.
Then, consider

$$\mathbb{B} = (1-\theta)\mathbb{I} + \theta\mathbb{A} \in \mathcal{L}_1$$

where $\mathbb{A} \in \mathcal{L}_1$. We have

$$\begin{aligned}
\|\mathbb{B}x - \mathbb{B}y\| =& \|(1-\theta)x + \theta\mathbb{A}x - (1-\theta)y + \theta\mathbb{A}y\| \\
\leq& (1-\theta)\|x-y\| + \theta\|\mathbb{A}x - \mathbb{A}y\| \\
\leq& (1-\theta)\|x-y\| + \theta\|x-y\| \\
\leq& \|x-y\|
\end{aligned}$$

which implies $\mathcal{N}_{\frac{1+R}{2}} \subseteq \mathcal{L}_1$.

(b) Let

$$\mathbb{A} = (1-\theta)\mathbb{I} + \theta\mathbb{B} \in \mathcal{N}_\theta$$

for some $\mathbb{B} \in \mathcal{L}_1$. Then,

$$\begin{aligned}\langle (\mathbb{I}-\mathbb{A})x - (\mathbb{I}-\mathbb{A})y, x-y\rangle =& \langle \theta x - \theta\mathbb{B}x - \theta y + \theta\mathbb{B}y, x-y\rangle \\ =& \theta\|x-y\|^2 - \theta\langle \mathbb{B}x - \mathbb{B}y, x-y\rangle.\end{aligned}$$

On the other hand, from $\mathbb{B} \in \mathcal{L}_1$, we have

$$\begin{aligned}\frac{1}{2\theta}\|(\mathbb{I}-\mathbb{A})x - (\mathbb{I}-\mathbb{A})y\|^2 =& \frac{1}{2\theta}\|\theta x - \theta\mathbb{B}x - \theta y + \theta\mathbb{B}y\|^2 \\ =& \frac{1}{2}\theta\|x-y\|^2 + \theta\langle \mathbb{B}x - \mathbb{B}y, x-y\rangle + \frac{1}{2}\theta\|\mathbb{B}x - \mathbb{B}y\|^2 \\ \leq& \frac{1}{2}\theta\|x-y\|^2 - \theta\langle \mathbb{B}x - \mathbb{B}y, x-y\rangle + \frac{1}{2}\theta\|x-y\|^2 \\ =& \theta\|x-y\|^2 - \theta\langle \mathbb{B}x - \mathbb{B}y, x-y\rangle \\ =& \langle (\mathbb{I}-\mathbb{A})x - (\mathbb{I}-\mathbb{A})y, x-y\rangle.\end{aligned}$$

This implies $\mathbb{A} \in \mathcal{N}_\theta \Rightarrow \mathbb{I}-\mathbb{A} \in \mathcal{C}_{1/(2\theta)}$.
Now lets show the converse. Suppose $\mathbb{I}-\mathbb{A} \in \mathcal{C}_{1/(2\theta)}$, which implies

$$\langle (\mathbb{I}-\mathbb{A})x - (\mathbb{I}-\mathbb{A})y, x-y\rangle \geq \frac{1}{2\theta}\|(\mathbb{I}-\mathbb{A})x - (\mathbb{I}-\mathbb{A})y\|^2.$$

If we decompose $\mathbb{A} = (1-\theta)\mathbb{I} + \theta\mathbb{B}$, then we have equivalent form

$$\begin{aligned}&\Leftrightarrow \langle \theta x - \theta\mathbb{B}x - \theta y + \theta\mathbb{B}y, x-y\rangle \geq \frac{1}{2\theta}\|\theta x - \theta\mathbb{B}x - \theta y + \theta\mathbb{B}y\|^2 \\ &\Leftrightarrow \theta\|x-y\|^2 - \theta\langle \mathbb{B}x - \mathbb{B}y, x-y\rangle \geq \frac{1}{2}\theta\|x-y\|^2 + \frac{1}{2}\theta\|\mathbb{B}x - \mathbb{B}y\|^2 - \theta\langle \mathbb{B}x - \mathbb{B}y, x-y\rangle \\ &\Leftrightarrow \theta\|x-y\|^2 - \theta\langle \mathbb{B}x - \mathbb{B}y, x-y\rangle \geq \frac{1}{2}\theta\|x-y\|^2 + \frac{1}{2}\theta\|\mathbb{B}x - \mathbb{B}y\|^2 - \theta\langle \mathbb{B}x - \mathbb{B}y, x-y\rangle \\ &\Leftrightarrow \frac{1}{2}\theta\|x-y\|^2 \geq \frac{1}{2}\theta\|\mathbb{B}x - \mathbb{B}y\|^2 \\ &\Leftrightarrow \|x-y\|^2 \geq \|\mathbb{B}x - \mathbb{B}y\|^2\end{aligned}$$

which implies $\mathbb{B} \in \mathcal{L}_1$, and finally $\mathbb{I}-\mathbb{A} \in \mathcal{C}_{1/(2\theta)} \Rightarrow \mathbb{A} \in \mathcal{N}_\theta$.

(c) (ii $\Leftrightarrow$ iii) is from part (b).
(i $\Leftrightarrow$ iii) is clear since for

$$\mathbb{A} = \frac{1}{2}\mathbb{I} + \frac{1}{2}\mathbb{B} \in \mathcal{N}_{1/2}$$

for some $\mathbb{B} \in \mathcal{L}_1$. We have

$$\mathbb{I}-\mathbb{A} = \frac{1}{2}\mathbb{I} + \frac{1}{2}(-\mathbb{B}) \in \mathcal{N}_{1/2}$$

and clearly $-\mathbb{B} \in \mathcal{L}_1$ as well. The converse is exactly the same.

(i $\Rightarrow$ iv)
For

$$\mathbb{A} = \frac{1}{2}\mathbb{I} + \frac{1}{2}\mathbb{B} \in \mathcal{N}_{1/2}$$

for some $\mathbb{B} \in \mathcal{L}_1$. We have

$$2\mathbb{A} - \mathbb{I} = \mathbb{B} \in \mathcal{L}_1.$$

(iv $\Rightarrow$ i)
For

$$2\mathbb{A} - \mathbb{I} = \mathbb{B} \in \mathcal{L}_1.$$

we have

$$\mathbb{A} = \frac{1}{2}\mathbb{I} + \frac{1}{2}\mathbb{B} \in \mathcal{N}_{1/2}.$$